

BRAINSTORM GROUP (BSG)

PHY113
(General Physics I)
MOCK TEST 1 [2021-22]



Federal University of Technology Minna

09068836600, 09068836661 *brainstormgroupfutmx001@gmail.com*

BRAINSTORM GROUP (BSG) PHY113 TEST (2021/2022)

1. The derived dimension $[ML^{-1}T^{-2}]$ is a dimension of
   A. Force B. Momentum
   C. Pressure D. Work

2. The derived dimension $[ML^2T^{-2}]$ is a dimension of
   A. Force B. Impulse
   C. Pressure D. Work

3. What are the correct dimensions for energy and force?
   A. ML^2T^{-2}, MLT^{-1} B. ML^2T^{-2}, MLT^{-2}
   C. $ML^{-3}T, MLT^2$ D. ML^2T^{-2}, ML^2T^2

4. What is the dimension of frequency?
   A. T B. T^{-2}
   C. T^2 D. T^{-1}

5. What is the dimension for surface tension?
   A. MT^{-2} B. MLT^{-3}
   C. $ML^{-2}T^2$ D. MLT^{-2}

6. The unit for tensile stress is the same as _____
   A. Force B. Tension
   C. Acceleration D. Pressure

7. What is the dimension of a, b and c respectively in the equation $S = at^2 + bt + c$
   A. $LT^{-1}, LT^{-2}, LT^{-3}$ B. $LT^{-3}, LT^{-2}, LT^{-1}$
   C. LT^2, LT^{-1}, LT D. LT^{-2}, LT^{-1}, L

8. Identify which of these quantities listed from (i) to (vii) are derived quantities:
   (i) mass (ii) length
   (iii) electric current (iv) density
   (v) temperature (vi) velocity
   (vii) energy
   A. (i), (ii), (iii) and (v)
   B. (iv), (v), (vi) and (v)
   C. (iv), (vi) and (vii)
   D. All of the above

9. The derived dimension $[ML^{-2}T^{-2}]$ is a dimension of (i) acceleration (ii) torque (iii) energy
   A. (i) only B. (iii) only
   C. (i) and (ii) only D. (ii) and (iii) only

10. Which of the following quantities has the same unit as the kilowatt – hour?
    A. $force \times acceleration$ B. $force \times velocity$
    C. $force \times velocity$ D. $force \times distance$

11. $0.1g/cm^3$ is equivalent to _____
    A. $10kg/m^3$ B. $100kg/m^3$
    C. $1000kg/m^3$ D. $10000kg/m^3$

12. The watt is equivalent to _____
    A. Nms^{-1} B. Js
    C. kgm^2s^2 D. Ns

13. Which of the following is not a unit of power?
    A. $Joule/second$ B. $ampere/volt$
    C. $ampere - volt$ D. $volt^2/\text{ohms}$

14. Which of the following is equivalent to kgm/s?
    A. J/s B. N/s
    C. Ns D. Nms

15. Convert the speed 108km/hr to units in m/s.
    A. 90m/s B. 45m/s
    C. 30m/s D. 60 m/s

16. Experiment shows that the frictional drag force F of a car moving in the air depends on the velocity v of its motion, density p of the air through which it travels, and the cross-sectional area of the car. If the force $F = kp^x v^{2y} a^z$, where x, y and z are integer and k is a dimensionless constant. Find the values of x, y, and z
    A. 1, 1, 1 B. 1, 1, −1
    C. 1, −1, 1 D. 1, 0, 1

17. Convert the speed 180km/hr to units in m/s.
    A. 100m/s B. 50m/s
    C. 150m/s D. 60 m/s

18. 20MPa is equal to _____
    A. $20N/m^2$ B. $20N/mm^2$
    C. $20kN/m^2$ D. $20N/cm^2$

19. A quantity with magnitude but no specified direction is _____ quantity
    A. vector B. scaler C. vectoral D. scalar

BRAINSTORM GROUP

20. Which of the following is a set of vectors?
A. weight, displacement and moment
B. velocity, volume and upthrust
C. density, capacitance and distance
D. mass, force and impulse

21. Which of the following is not a fundamental S.I. unit?
A. meter B. ampere
C. mole D. newton

22. A girl walks 12m northwards, 5km eastwards and 7m southwards. Her total displacement is
A. 5m, North B. 5m, East
C. 7.07m, S45^{0}W D. 7.07m, N45^{0}E

23. Two forces, 12N and 16N inclined at an angle to each other, have a resultant which is parallel to the 16N force. The value of cos is
A. 1 B. $\frac{3}{4}$
C. $\frac{1}{2}$ D. 0

24. A body of mass M rests on a plane inclined at an angle α to the horizontal. The component of the weight of the body along he normal to the plane is
A. $Mgsin\alpha$ B. $Mgcos\alpha$
C. $Mgtan\alpha$ D. $Mgcot\alpha$

25. A car travels 20.0km due north and then 35.0km in a direction 60^0 west of north. Find the magnitude and direction of the car's resultant displacement.
A. 48.2km 39^0 W of N B. 53.3km 39^0 W of N
C. 48.2km 51^0 W of N D. 53.3km 51^0 W of N

26. Can the magnitude of Cristiano Ronaldo's displacement on the pitch be greater than the distance he travelled?
A. Yes B. No C. Maybe
D. It depends on the ball's position

27. Two vectors are given by A = 3i − 2j and B = −i − j, calculate the value of A + B.
A. 4i + 3j B. 2i − 3j
C. 2i − 4j D. −3i + 2j

28. A runner makes one lap around a track of radius 100m in a time of 25s. What is the runner's displacement?
A. 200πm B. 200m
C. 100m D. 0m

29. A runner makes one lap around a track of radius 100m in a time of 25s. What is the distance covered by the runner?
A. 200πm B. 200m
C. 100m D. 0m

30. A force 2500N acting at 60^0 to the vertical is used to pull a lorry, what is the amount of force trying to lift the lorry
A. 1250N B. 2165N
C. 4330N D. 2065N

31. The resultant of two forces is the diagonal of the forces from their point of action, this the
A. Triangular law B. Lami's theorem
C. Polygon law D. Parallelogram law

32. Which of the following is not a scalar quantity?
A. Electric Potential B. Tensile Stress
C. Surface Tension D. Young Modulus

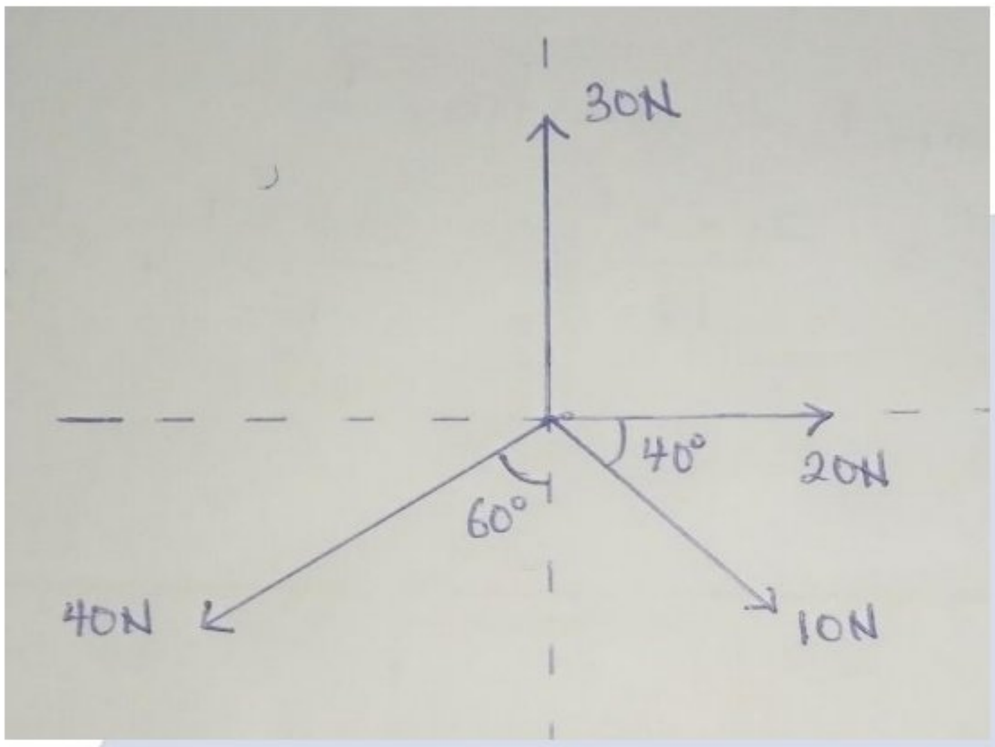
33. Two perpendicular forces have a resultant of 13N. If one of the forces is 5N, the other force is
A. 8N B. 9N
C. 12N D. 14N

34. Two forces will have their resultant to be minimum if the angle between them is
A. 0^0 B. 45^0
C. 90^0 D. 180^0

35. The velocity v of the wave set up by plucking a stretched string is found to depend on the tension T on the string, its length l and its mass m, and is given by $v = kT^x l^y m^z$ where x, y, z are unknown numbers and k is a constant. Find the value of the summation of (x + y).
A. $\frac{1}{2}$ B. 1
C. $\frac{3}{2}$ D. 2

BRAINSTORM GROUP

Use the below diagram to answer questions 36 – 39.



36. What is the net horizontal force?
    A. -6.98N
    B. 62.30N
    C. -4.64N
    D. 23.02N

37. What is the net vertical force?
    A. 56.42N
    B. 43.57N
    C. 23.57N
    D. 3.57N

38. What is the resultant force?
    A. 85.05N
    B. 48.95N
    C. 4.44N
    D. 32.95N

39. What is the resultant angle from the positive horizontal?
    A. 27^0
    B. 207^0
    C. 63^0
    D. 153^0

40. This test was organized by _____
    A. Brainstorming Group
    B. Brainstormers Group
    C. Brainstorm Group
    D. Brainbox Group
    E. Brainstorm's Group

BRAINSTORM GROUP

ANSWERS							
1.	C	11.	B	21.	D	31.	D
2.	D	12.	A	22.	D	32.	B
3.	B	13.	B	23.	D	33.	C
4.	D	14.	C	24.	B	34.	D
5.	A	15.	C	25.	A	35.	B
6.	D	16.	A	26.	B	36.	A
7.	D	17.	B	27.	B	37.	D
8.	C	18.	B	28.	D	38.	C
9.	D	19.	D	29.	D	39.	D
10.	C	20.	A	30.	A	40.	C

BRAINSTORM GROUP